Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/03/2006
PI Name:	Jin, J.-P. M.D., Ph.D.		
Project Title:	Adaptation Of Myocardial Contractile Proteins In Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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City:	Evanston	**State:**	IL
Zip Code:	60201-1718	**Congressional District:**	9
Comments:	Former affiliation: Case Western Reserve University; moved in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: NCE to 6/30/2008 per A. Chu (4/07) NOTE: NCE to 6/30/07 per A. Chu (7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK62G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), an essential protein in the Ca2+-regulatory system of cardiac muscle. This novel structural regulation of cTnI removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA). Beta-adrenergic-regulated PKA phosphorylation of cTnI reduces the sensitivity of cardiac muscle to Ca2+. Therefore, the up-regulation of N-terminal truncated cTnI may provide a compensation for cardiac function against the microgravity-induced reduction of contractility. This research proposal describes ground based studies to investigate the role of the proteolytic regulation of cTnI in adaptation to microgravity for potential use in the prevention and treatment of cardiovascular deconditioning during and after space flight. We plan to study the functional changes in cTnI caused by the N-terminal truncation and the effect on cardiac muscle contraction in simulated microgravity. Conformation and function of the N-terminal truncated cTnI will be investigated in vitro and in the heart of transgenic mice. We shall examine the effect of the removal of PKA phosphorylation sites from cTnI on beta-adrenergic regulation during cardiac adaptation in simulated microgravity. We shall also investigate the regulatory mechanisms for this proteolytic modification of cTnI during adaptation to microgravity. We shall focus on the role of calpain that is a family of regulatory proteases and known to function in the degradation of troponin subunits. The role of PKA and PKC phosphorylation of cTnI in the regulation of sensitivity to calpain cleavage will be characterized. This research will provide new information for understanding cardiac adaptation in microgravity and protecting cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our study will contribute to both space life science and human life quality on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our discovery that the N-terminal modification of cardiac troponin I enhances cardiac function has lead to an approach to the development of new drugs for the treatment of heart failure that is the most important death cause for cardiovascular diseases.
Task Progress:	Progress in the reporting period 1) Characterization of the function of N-terminal truncated cardiac troponin I: Using transgenic mice that we developed in this project to over-express the N-terminal truncated cardiac troponin I (cTnI-ND) in the heart (Barbato, J.C., Huang, Q.-Q., Hossain, M.M., Bond, M., and Jin, J.-P. J. Biol. Chem. 280:6602-9, 2005), we are performing an in vivo functional characterization by echocardiography. The initial results show an improved cardiac function of the transgenic mice, especially significant at the old age. Experiments are underway to further investigate this novel observation. 2) Mechanisms that regulate the proteolytic N-terminal truncation of cardiac troponin I: To investigate the physiological regulation of the cardiac troponin I N-terminal truncation and the role of PKA phosphorylation under beta-adrenergic regulation, we found increases of N-terminal truncation of cardiac troponin I in a Gs alpha conditional knock out mouse model. The disruption of Gs alpha signaling pathway results in diminished PKA activity. The corresponding up-regulation of N-terminal truncated cardiac troponin I supports the hypothesis that this proteolytic regulation is a part of the physiological regulation of cardiac function. In collaboration with NIH/NIDDK, we are currently performing a microarray examination of the Gs alpha knockout mouse heart to investigate potential gene regulations linking to the enhanced N-terminal truncation of cardiac troponin I. Since the Gs alpha knock out mice appear to have a decreased potential of cardiac function (unpublished observation), we are testing a hypothesis that the N-terminal truncation of cardiac troponin I has a compensatory role, consistent with its potential use as a countermeasure of the microgravity-induced decrease of cardiac function. 3) In addition to the development of countermeasures to the decreased ventricular filling during long space flights, our finding that N-terminal truncation of cardiac troponin I is an effective novel mechanism to enhance myocardial relaxation has an attractive application on earth as a new target for the treatment of heart failure. We have submitted a full US patent application for this application. 4) To investigate the role of PKA and PKC phosphorylation in regulating the proteolytic modification of cardiac troponin I, we have carried out experiments that demonstrate a) the specific cleavage of cardiac TnI N-terminal domain is dependent on the myofibril structure; b) PKC phosphorylation may be a target for promoting this cleavage as a countermeasure to improve the cardiac function of astronauts during long space flight; and c) the effect of inhibition of A-kinase anchoring proteins (AKAPs) on increasing the N-terminal truncation of cardiac troponin I. (An abstract has been accepted by the Heart Failure Society of America (HFSA) annual meeting: B. McConnell, N. Mal, F. Forudi, J. Bautista, J.-P. Jin, Z. Popovic, M. Penn, Targeted Disruption of PKA Binding to AKAPs Increases Cardiac Contractility and Decreases PKA-Dependent Phosphorylation, in vivo.)
Bibliography Type:	Description: (Last Updated: 04/20/2009)
Abstracts for Journals and Proceedings	McConnell B, Mal N, Forudi F, Bautista J, Jin J-P, Popovic Z, Penn M. "Targeted disruption of PKA binding to AKAPs increases cardiac contractility and decreases PKA-dependent phosphorylation, in vivo." 10th Annual Scientific Meeting Heart Failure Society of America, Seattle, WA, September 10-13, 2006. 10th Annual Scientific Meeting Heart Failure Society of America, September 2006. , Sep-2006